

Stats 652 - Homework 1

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1. (0 points) **Problem 1.1.**

Define

$$g(\mu, \theta) = e^{-\mu^2\theta} + e^{-\theta} + \frac{1}{3} \log \theta$$

on

$$C = \{(\mu, \theta) : 0 < \theta < \infty, \mu > \theta - 1\}.$$

Show that g has only one critical point in C , and that it is a local maximum but not a global maximum.

Solution: We first find all critical points of g that lie in C . The first-order partial derivatives are

$$\frac{\partial g}{\partial \mu} = -2\mu\theta e^{-\mu^2\theta}$$

and

$$\frac{\partial g}{\partial \theta} = -\mu^2 e^{-\mu^2\theta} - e^{-\theta} + \frac{1}{3\theta}.$$

Because $\theta > 0$ and $e^{-\mu^2\theta} > 0$, the equation

$$\frac{\partial g}{\partial \mu} = 0$$

holds if and only if

$$\mu = 0.$$

Therefore every critical point of g must have the form $(0, \theta)$.

Substituting $\mu = 0$ into the second score equation gives

$$-e^{-\theta} + \frac{1}{3\theta} = 0,$$

or equivalently,

$$e^{\theta} = 3\theta.$$

Define

$$h(\theta) = e^\theta - 3\theta, \quad \theta > 0.$$

Then

$$h'(\theta) = e^\theta - 3$$

and

$$h''(\theta) = e^\theta > 0.$$

Hence h is strictly convex on $(0, \infty)$ and has a unique minimum at

$$\theta = \log 3.$$

Moreover,

$$h(0) = 1 > 0, \quad h(1) = e - 3 < 0,$$

so the intermediate value theorem implies that h has a root in $(0, 1)$. Also,

$$h(\log 3) = 3 - 3 \log 3 < 0,$$

while $h(\theta) \rightarrow \infty$ as $\theta \rightarrow \infty$. Therefore h has another root in $(\log 3, \infty)$. Since h is strictly convex and has only one minimum, these are its only two positive roots.

However, a point of the form $(0, \theta)$ lies in C only when

$$0 > \theta - 1,$$

which is equivalent to

$$\theta < 1.$$

Thus the larger root does not give a point in C . Let $\theta_0 \in (0, 1)$ denote the unique solution of

$$e^{\theta_0} = 3\theta_0.$$

Then

$$(0, \theta_0)$$

is the unique critical point of g in C .

We now classify this critical point. The relevant second derivatives are

$$g_{\mu\mu}(\mu, \theta) = -2\theta e^{-\mu^2\theta} + 4\mu^2\theta^2 e^{-\mu^2\theta},$$

$$g_{\mu\theta}(\mu, \theta) = -2\mu e^{-\mu^2\theta} + 2\mu^3\theta e^{-\mu^2\theta},$$

and

$$g_{\theta\theta}(\mu, \theta) = \mu^4 e^{-\mu^2\theta} + e^{-\theta} - \frac{1}{3\theta^2}.$$

At $(0, \theta_0)$, these reduce to

$$g_{\mu\mu}(0, \theta_0) = -2\theta_0 < 0, \quad g_{\mu\theta}(0, \theta_0) = 0.$$

Because θ_0 satisfies

$$e^{-\theta_0} = \frac{1}{3\theta_0},$$

we also have

$$g_{\theta\theta}(0, \theta_0) = \frac{1}{3\theta_0} - \frac{1}{3\theta_0^2} = \frac{\theta_0 - 1}{3\theta_0^2} < 0,$$

where the final inequality follows from $\theta_0 < 1$.

Therefore the Hessian at the critical point is

$$\nabla^2 g(0, \theta_0) = \begin{pmatrix} -2\theta_0 & 0 \\ 0 & \frac{\theta_0 - 1}{3\theta_0^2} \end{pmatrix}.$$

Both diagonal entries are negative, so the Hessian is negative definite. Hence $(0, \theta_0)$ is a strict local maximum.

Finally, this local maximum is not a global maximum. Consider the sequence

$$(\mu, \theta) = (\theta, \theta), \quad \theta > 0.$$

Each such point lies in C , since

$$\mu = \theta > \theta - 1.$$

Along this sequence,

$$g(\theta, \theta) = e^{-\theta^3} + e^{-\theta} + \frac{1}{3} \log \theta.$$

As $\theta \rightarrow \infty$, the first two terms tend to zero, whereas

$$\frac{1}{3} \log \theta \rightarrow \infty.$$

Consequently,

$$g(\theta, \theta) \rightarrow \infty.$$

Thus g is unbounded above on C , and its unique critical point is a strict local maximum but not a global maximum.

2. (0 points) **Problem 1.2.**

Let $X_1, \dots, X_n$ be a random sample from a $\mathcal{N}(\mu, \sigma^2)$ distribution, where

$$\mu \in \mathbb{R} \quad \text{and} \quad \sigma^2 > 0.$$

- (a) Show that the likelihood goes to zero as (μ, σ^2) approaches the boundary of the parameter space.

- (b) Show that the likelihood is not strictly log-concave in (μ, σ^2) .
(c) Verify directly that the log likelihood is strictly concave in (η_1, η_2) , where

$$\eta_1 = \frac{1}{\sigma^2}, \quad \eta_2 = \frac{\mu}{\sigma^2}.$$

Solution: Let

$$v = \sigma^2, \quad Q(\mu) = \sum_{i=1}^n (x_i - \mu)^2, \quad S = \sum_{i=1}^n (x_i - \bar{x})^2.$$

Then

$$Q(\mu) = S + n(\mu - \bar{x})^2,$$

so the likelihood is

$$L(\mu, v) = (2\pi v)^{-n/2} \exp \left\{ -\frac{Q(\mu)}{2v} \right\}.$$

- (a) Assume $S > 0$, which holds with probability one under the normal model when $n \geq 2$. Since $Q(\mu) \geq S$,

$$0 \leq L(\mu, v) \leq (2\pi v)^{-n/2} \exp \left\{ -\frac{S}{2v} \right\}.$$

The right-hand side tends to 0 as $v \downarrow 0$, uniformly in μ . Also,

$$L(\mu, v) \leq (2\pi v)^{-n/2} \longrightarrow 0 \quad \text{as } v \rightarrow \infty.$$

Finally, for fixed μ , $L(\mu, v)$ is maximized over $v > 0$ at $v = Q(\mu)/n$, and hence

$$\sup_{v>0} L(\mu, v) = \left(\frac{n}{2\pi e Q(\mu)} \right)^{n/2} \longrightarrow 0 \quad \text{as } |\mu| \rightarrow \infty,$$

because $Q(\mu) = S + n(\mu - \bar{x})^2 \rightarrow \infty$. Therefore the likelihood tends to zero along every sequence leaving compact subsets of $\mathbb{R} \times (0, \infty)$. The assumption $S > 0$ is necessary: if all observations equal x_0 , then $L(x_0, v) = (2\pi v)^{-n/2} \rightarrow \infty$ as $v \downarrow 0$.

- (b) Ignoring constants that do not depend on (μ, v) ,

$$\ell(\mu, v) = -\frac{n}{2} \log v - \frac{Q(\mu)}{2v}.$$

Since

$$Q'(\mu) = 2n(\mu - \bar{x}),$$

we have

$$\frac{\partial \ell}{\partial \mu} = -\frac{n(\mu - \bar{x})}{v}, \quad \frac{\partial \ell}{\partial v} = -\frac{n}{2v} + \frac{Q(\mu)}{2v^2}.$$

The Hessian is

$$H_{\mu,v} = \begin{pmatrix} -\frac{n}{v} & \frac{n(\mu - \bar{x})}{v^2} \\ \frac{n(\mu - \bar{x})}{v^2} & \frac{n}{2v^2} - \frac{Q(\mu)}{v^3} \end{pmatrix}.$$

Its determinant is

$$\begin{aligned} \det H_{\mu,v} &= -\frac{n}{v} \left(\frac{n}{2v^2} - \frac{Q(\mu)}{v^3} \right) - \frac{n^2(\mu - \bar{x})^2}{v^4} \\ &= \frac{n}{v^4} \left[Q(\mu) - n(\mu - \bar{x})^2 - \frac{nv}{2} \right] \\ &= \frac{n}{v^4} \left(S - \frac{nv}{2} \right). \end{aligned}$$

If

$$v > \frac{2S}{n},$$

then

$$\det H_{\mu,v} < 0.$$

So the Hessian is indefinite at those points. Therefore the log likelihood is not concave in (μ, v) , and hence it is not strictly log-concave in (μ, σ^2) .

(c) Let

$$\eta_1 = \frac{1}{v}, \quad \eta_2 = \frac{\mu}{v}.$$

Then

$$v = \frac{1}{\eta_1}, \quad \mu = \frac{\eta_2}{\eta_1},$$

and the parameter space is

$$\{(\eta_1, \eta_2) : \eta_1 > 0, \eta_2 \in \mathbb{R}\}.$$

Expanding the quadratic term gives

$$\sum_{i=1}^n (x_i - \mu)^2 = \sum_{i=1}^n x_i^2 - 2\mu \sum_{i=1}^n x_i + n\mu^2.$$

Therefore,

$$\ell(\eta_1, \eta_2) = C + \frac{n}{2} \log \eta_1 - \frac{\eta_1}{2} \sum_{i=1}^n x_i^2 + \eta_2 \sum_{i=1}^n x_i - \frac{n\eta_2^2}{2\eta_1}.$$

The first derivatives are

$$\frac{\partial \ell}{\partial \eta_1} = \frac{n}{2\eta_1} - \frac{1}{2} \sum_{i=1}^n x_i^2 + \frac{n\eta_2^2}{2\eta_1^2},$$

and

$$\frac{\partial \ell}{\partial \eta_2} = \sum_{i=1}^n x_i - \frac{n\eta_2}{\eta_1}.$$

Thus,

$$H_\eta = \begin{pmatrix} -\frac{n}{2\eta_1^2} - \frac{n\eta_2^2}{\eta_1^3} & \frac{n\eta_2}{\eta_1^2} \\ \frac{n\eta_2}{\eta_1^2} & -\frac{n}{\eta_1} \end{pmatrix}.$$

The upper-left entry is negative:

$$-\frac{n}{2\eta_1^2} - \frac{n\eta_2^2}{\eta_1^3} < 0.$$

Also,

$$\begin{aligned} \det H_\eta &= \left(-\frac{n}{2\eta_1^2} - \frac{n\eta_2^2}{\eta_1^3} \right) \left(-\frac{n}{\eta_1} \right) - \left(\frac{n\eta_2}{\eta_1^2} \right)^2 \\ &= \frac{n^2}{2\eta_1^3} > 0. \end{aligned}$$

Therefore H_η is negative definite for every $\eta_1 > 0$. Hence the log likelihood is strictly concave in (η_1, η_2) .

3. (0 points) **Problem 1.3.**

- (a) Show that if $f(x; \theta)$ is an exponential family in $\theta = (\theta_1, \theta_2)$, then, for fixed θ_2 , it is an exponential family in θ_1 .
- (b) Show whether, if every coordinate slice is an exponential family, the joint family must be an exponential family. (Optional)

Solution:

(a) Suppose

$$f(x; \theta_1, \theta_2) = \exp \left\{ \sum_{j=1}^k c_j(\theta_1, \theta_2) T_j(x) + d(\theta_1, \theta_2) + S(x) \right\} I_A(x),$$

where A does not depend on (θ_1, θ_2) .

Fix $\theta_2 = \theta_2^0$. Then

$$f(x; \theta_1, \theta_2^0) = \exp \left\{ \sum_{j=1}^k c_j(\theta_1, \theta_2^0) T_j(x) + d(\theta_1, \theta_2^0) + S(x) \right\} I_A(x).$$

This is an exponential-family representation in θ_1 . Therefore, for fixed θ_2 , the family is an exponential family in θ_1 .

(b) No.

Let

$$\mathcal{X} = A = \mathbb{N}, \quad h(x) = 2^{-x}, \quad x \in \mathbb{N}.$$

For each $j \in \mathbb{N}$, define

$$U(x) = 4^{-x}, \quad V(x) = 3^{-x}, \quad T_j(x) = \mathbf{1}\{x = j\}.$$

Let the parameter space be $\Theta = \mathbb{N}^2$. For $(m, r) \in \Theta$, define

$$Z_{m,r} = \sum_{x \in \mathbb{N}} h(x) \exp \{mU(x) + rV(x) + \mathbf{1}\{m = r\}T_m(x)\},$$

and set

$$f(x; m, r) = \exp \{mU(x) + rV(x) + \mathbf{1}\{m = r\}T_m(x) - \log Z_{m,r} + \log h(x)\} I_{\mathbb{N}}(x).$$

Each $f(\cdot; m, r)$ is a pmf on $\mathbb{N}$, since $0 < Z_{m,r} < \infty$.

We first record the following linear-independence fact, which we use below. Suppose

$$a_0 + aU(x) + bV(x) + \sum_{j \in J} b_j T_j(x) = 0 \quad \text{for all } x \in \mathbb{N},$$

where J is finite. If $x > \max J$, then all the $T_j(x)$ vanish, so

$$a_0 + a4^{-x} + b3^{-x} = 0 \quad \text{for all sufficiently large } x.$$

Letting $x \rightarrow \infty$ gives $a_0 = 0$. Since 4^{-x} and 3^{-x} are linearly independent as functions of x (their ratio $(4/3)^{-x}$ is non-constant, so no nontrivial combination can vanish at two or more values of x), this also gives $a = b = 0$. Evaluating at $x = j$ then shows $b_j = 0$ for each $j \in J$.

Thus the functions

$$1, U, V, T_1, T_2, \dots$$

are linearly independent in every finite combination.

Now fix $r = r_0$. Then

$$f(x; m, r_0) = \exp \{mU(x) + \mathbf{1}\{m = r_0\}T_{r_0}(x) - \log Z_{m, r_0} + S_{r_0}(x)\} I_{\mathbb{N}}(x),$$

where

$$S_{r_0}(x) = r_0 V(x) + \log h(x).$$

So for fixed r_0 , this is an exponential family in m , with statistics $U(x)$ and $T_{r_0}(x)$.

Likewise, if $m = m_0$ is fixed, then

$$f(x; m_0, r) = \exp \{rV(x) + \mathbf{1}\{r = m_0\}T_{m_0}(x) - \log Z_{m_0, r} + S_{m_0}(x)\} I_{\mathbb{N}}(x),$$

where

$$S_{m_0}(x) = m_0 U(x) + \log h(x).$$

So every slice with m fixed is also an exponential family.

Suppose now that the full family were a k -dimensional exponential family for some finite k . Then all log-density ratios would lie in one finite-dimensional vector space spanned by 1 and the fixed statistics of that representation.

Comparing the off-diagonal points $(2, 4)$ and $(1, 4)$ gives

$$\log \frac{f(x; 2, 4)}{f(x; 1, 4)} = U(x) + \text{constant},$$

so U lies in that finite-dimensional space. Comparing $(4, 2)$ and $(4, 1)$ gives

$$\log \frac{f(x; 4, 2)}{f(x; 4, 1)} = V(x) + \text{constant},$$

so V also lies in that space.

Finally, for each $j \in \mathbb{N}$,

$$\log \frac{f(x; j, j)}{f(x; j, j+1)} = T_j(x) - V(x) + \text{constant}.$$

Since V is already in the space, it follows that every T_j is also in the same finite-dimensional space. But $T_1, T_2, \dots$ are linearly independent, which is impossible in a finite-dimensional space. This contradiction shows that the joint family is not a finite-dimensional exponential family.

4. (0 points) **Problem 1.4.** (Optional)

Show that a k -dimensional exponential family is in minimal representation form if and only if

- (a) $\sum_{j=1}^k a_j c_j(\theta) = a_0$ for all θ implies $a_0 = a_1 = \dots = a_k = 0$;
(b) $\sum_{j=1}^k b_j T_j(x) = b_0$ for all $x \in A$ implies $b_0 = b_1 = \dots = b_k = 0$.

Solution: Write

$$f(x; \theta) = \exp\{c(\theta)^\top T(x) + d(\theta) + S(x)\} I_A(x).$$

Suppose first that the representation is minimal. If $a^\top c(\theta) = a_0$ for a nonzero vector a , choose an invertible linear transformation whose last transformed coefficient is $a^\top c(\theta)$. That coefficient is constant, so its term is independent of θ and can be absorbed into $S(x)$. This removes one statistic and contradicts minimality. Hence (a) holds.

Likewise, if $b^\top T(x) = b_0$ for a nonzero vector b , choose an invertible linear transformation whose last transformed statistic is $b^\top T(x)$. Since this statistic is constant on A , its term can be absorbed into $d(\theta)$. This again reduces the number of statistics, contradicting minimality. Hence (b) holds.

Conversely, assume (a) and (b). Fix $\theta_0 \in \Theta$, $x_0 \in A$, and define

$$K(\theta, x) = \log f(x; \theta) - \log f(x_0; \theta) - \log f(x; \theta_0) + \log f(x_0; \theta_0).$$

This quantity is defined directly from the density f , so it does not depend on which representation is used to compute it. Substituting $f(x; \theta) = \exp\{c(\theta)^\top T(x) + d(\theta) + S(x)\} I_A(x)$, the terms $d(\theta)$ and $S(x)$ cancel in the double difference, leaving

$$K(\theta, x) = [c(\theta) - c(\theta_0)]^\top [T(x) - T(x_0)].$$

Condition (a) implies that $\{c(\theta) - c(\theta_0) : \theta \in \Theta\}$ spans $\mathbb{R}^k$ (otherwise a nonzero vector orthogonal to this span would give a forbidden affine relation among the c_j), and condition (b) similarly implies $\{T(x) - T(x_0) : x \in A\}$ spans $\mathbb{R}^k$. We may therefore choose $\theta_1, \dots, \theta_k \in \Theta$ and $x_1, \dots, x_k \in A$ so that the $k \times k$ matrix $[K(\theta_i, x_j)]_{i,j=1}^k$ has rank k .

Now suppose some representation of the same family used only $r < k$ statistics, say $f(x; \theta) = \exp\{\sum_{l=1}^r c'_l(\theta) T'_l(x) + d'(\theta) + S'(x)\} I_A(x)$. Because K depends only on f , applying the same double-difference identity to this representation gives

$$K(\theta, x) = \sum_{l=1}^r [c'_l(\theta) - c'_l(\theta_0)] [T'_l(x) - T'_l(x_0)].$$

Evaluating at the chosen grid, $[K(\theta_i, x_j)] = \sum_{l=1}^r u_l v_l^\top$ where $u_l = (c'_l(\theta_i) - c'_l(\theta_0))_{i=1}^k$ and $v_l = (T'_l(x_j) - T'_l(x_0))_{j=1}^k$. A sum of r rank-one matrices has rank at most r , since its column space is spanned by $u_1, \dots, u_r$. But this matrix was constructed to have rank k , so $r \geq k$. Hence no representation with fewer than k statistics exists, so the given representation is minimal.

5. (0 points) **Problem 1.5.**

Let

$$f(x; \mu) = \frac{2}{\Gamma(1/4)} e^{-(x-\mu)^4}, \quad x, \mu \in \mathbb{R}.$$

- (a) Show that this is a k -dimensional curved exponential family and specify k .
- (b) For a random sample of size n , determine the roots of the likelihood equation and the MLE.
- (c) Find a minimal sufficient statistic for μ .

Solution:

- (a) Expanding the exponent,

$$-(x - \mu)^4 = -x^4 + 4\mu x^3 - 6\mu^2 x^2 + 4\mu^3 x - \mu^4.$$

Thus

$$f(x; \mu) = \exp \left\{ 4\mu^3 x - 6\mu^2 x^2 + 4\mu x^3 - \mu^4 - x^4 + \log \frac{2}{\Gamma(1/4)} \right\} I_{\mathbb{R}}(x).$$

Take

$$T(x) = (x, x^2, x^3)^\top, \quad c(\mu) = (4\mu^3, -6\mu^2, 4\mu)^\top.$$

The functions $1, x, x^2, x^3$ are linearly independent, as are the functions $1, \mu, \mu^2, \mu^3$. Hence the representation is minimal and $k = 3$. Since $\{c(\mu) : \mu \in \mathbb{R}\}$ is a one-dimensional nonlinear curve in $\mathbb{R}^3$ and does not contain an open subset of $\mathbb{R}^3$, this is a curved exponential family.

- (b) Ignoring constants,

$$\ell(\mu) = - \sum_{i=1}^n (x_i - \mu)^4, \quad \ell'(\mu) = 4 \sum_{i=1}^n (x_i - \mu)^3.$$

Let

$$F(\mu) = \sum_{i=1}^n (x_i - \mu)^3.$$

Each summand is strictly decreasing in μ , so F is strictly decreasing. Also,

$$F(\mu) \rightarrow \infty \quad (\mu \rightarrow -\infty), \quad F(\mu) \rightarrow -\infty \quad (\mu \rightarrow \infty).$$

Therefore the likelihood equation

$$\sum_{i=1}^n (x_i - \mu)^3 = 0$$

has exactly one root. Since $\ell(\mu) \rightarrow -\infty$ as $|\mu| \rightarrow \infty$, that root is the unique global MLE.

(c) The factorization theorem gives sufficiency of

$$T(X) = \left(\sum_{i=1}^n X_i, \sum_{i=1}^n X_i^2, \sum_{i=1}^n X_i^3 \right).$$

For sample points $x, y \in \mathbb{R}^n$, the log likelihood ratio is a polynomial in μ . It is independent of μ exactly when the coefficients of μ , μ^2 , and μ^3 agree, which is equivalent to $T(x) = T(y)$. The likelihood-ratio criterion therefore shows that T is minimal sufficient.

6. (0 points) **Problem 1.6.**

Show that if η represents the natural parameters of an exponential family, then η^* is also a natural parametrization if and only if

$$\eta^* = A\eta + a$$

for some nonsingular matrix A and some $a \in \mathbb{R}^k$.

Solution: Throughout, “natural parametrization” is taken to mean a minimal (full-rank) representation, as is standard; this guarantees η and η^* have the same dimension k (the family’s minimal order) and lets us invoke the characterization of minimality from Problem 1.4.

Assume both are minimal full k -dimensional natural representations of the same family $\{f_\theta : \theta \in \Theta\}$, related to the original parameter θ by bijections $\eta = \eta(\theta)$ and $\eta^* = \eta^*(\theta)$. Since both maps are invertible, η^* is in turn a well-defined function of η alone, namely $\eta^*(\eta) := \eta^*(\theta(\eta))$; this is the map we show to be affine. In the first representation,

$$f_\eta(x) = \exp\{\eta^\top T(x) + d_0(\eta) + S(x)\} I_A(x).$$

By minimality, we can choose $x_0, x_1, \dots, x_k \in A$ so that the matrix D whose i th row is $(T(x_i) - T(x_0))^\top$ is nonsingular. Writing out the log-density ratio at each x_i against x_0 ,

$$\log \frac{f_\eta(x_i)}{f_\eta(x_0)} = \eta^\top (T(x_i) - T(x_0)) + (S(x_i) - S(x_0)),$$

the $d_0(\eta)$ terms cancel because they do not depend on x . Hence, setting $b_i := S(x_i) - S(x_0)$ (which does not depend on η) and $b := (b_1, \dots, b_k)^\top$, the vector of log-density ratios has the form

$$r(\eta) = \left(\log \frac{f_\eta(x_i)}{f_\eta(x_0)} \right)_{i=1}^k = b + D\eta.$$

Under the other minimal natural representation, with carrier function S^* , the identical computation gives

$$r(\eta) = b^* + D^* \eta^*(\eta), \quad b_i^* := S^*(x_i) - S^*(x_0),$$

where D^* is the (also nonsingular, by minimality of the second representation) matrix with i th row $(T^*(x_i) - T^*(x_0))^\top$. Since $r(\eta)$ is the same quantity computed from the same density f_θ in each representation, the two expressions agree, so

$$\eta^*(\eta) = (D^*)^{-1} D \eta + (D^*)^{-1} (b - b^*) = A \eta + a,$$

and $A = (D^*)^{-1} D$ is nonsingular.

Conversely, suppose $\eta^* = A \eta + a$ with A nonsingular. Then

$$\eta = A^{-1}(\eta^* - a)$$

and

$$\eta^\top T(x) = (\eta^*)^\top A^{-\top} T(x) - a^\top A^{-\top} T(x).$$

Set

$$T^*(x) = A^{-\top} T(x), \quad S^*(x) = S(x) - a^\top A^{-\top} T(x).$$

The second term is absorbed into the carrier S^* , while the first is $(\eta^*)^\top T^*(x)$. Hence η^* is also a natural parametrization.

7. (0 points) **Problem 1.7.**

For a k -dimensional exponential family in natural form, prove the convexity, derivative, Hessian, and MGF identities stated in the assignment.

Solution: Write the natural form as

$$f_\eta(x) = \exp\{\eta^\top T(x) + d_0(\eta) + S(x)\} I_A(x).$$

Let $h(x) = e^{S(x)} I_A(x)$ and define

$$Z(\eta) = e^{-d_0(\eta)} = \int_A e^{\eta^\top T(x)} h(x) d\nu(x).$$

Assume the representation is minimal and work in the interior of the natural parameter space, where differentiation under the integral is valid.

For any $a \neq 0$,

$$a^\top \nabla^2 Z(\eta) a = \int_A (a^\top T(x))^2 e^{\eta^\top T(x)} h(x) d\nu(x) > 0.$$

Minimality rules out $a^\top T(x)$ being almost surely constant. Thus $Z(\eta) = e^{-d_0(\eta)}$ is strictly convex.

Now let

$$A(\eta) = \log Z(\eta) = -d_0(\eta).$$

Differentiating gives

$$\nabla A(\eta) = \mathbb{E}_\eta[T(X)]$$

and

$$\nabla^2 A(\eta) = \text{Cov}_\eta(T(X)).$$

For $a \neq 0$,

$$a^\top \nabla^2 A(\eta) a = \text{Var}_\eta(a^\top T(X)) > 0,$$

so A is strictly convex and $d_0 = -A$ is strictly concave. In particular,

$$\nabla d_0(\eta) = -\mathbb{E}_\eta[T(X)]$$

and

$$\frac{\partial^2 d_0(\eta)}{\partial \eta_i \partial \eta_j} = -\text{Cov}_\eta(T_i(X), T_j(X)).$$

For one observation, the data-dependent term $\eta^\top T(x)$ is linear in η . Therefore the Hessian of the log likelihood is

$$H_\ell(\eta) = \nabla^2 \ell(\eta) = -\text{Cov}_\eta(T(X)),$$

so

$$a^\top H_\ell(\eta) a = -\text{Var}_\eta \left(\sum_{j=1}^k a_j T_j(X) \right).$$

Finally, whenever $\eta + t$ belongs to the natural parameter space,

$$\begin{aligned} M_{T,\eta}(t) &= \mathbb{E}_\eta \left[e^{t^\top T(X)} \right] \\ &= e^{d_0(\eta)} \int_A e^{(\eta+t)^\top T(x)} h(x) d\nu(x) \\ &= \frac{Z(\eta+t)}{Z(\eta)} = \exp\{d_0(\eta) - d_0(\eta+t)\}. \end{aligned}$$

8. (0 points) **Problem 1.8.**

Find the MLEs, if they exist, in the following cases.

- (a) $X_1 \sim \mathcal{N}(\mu, \sigma^2)$ and $X_2 \sim \mathcal{N}(\mu, 1)$ independently.
- (b) X_i iid $\mathcal{N}(\theta, \theta^2)$.

- (c) X_i iid $\mathcal{N}(\theta, \theta)$, $\theta > 0$.
- (d) X_i iid $\text{Gamma}(\theta, \theta)$, $\theta > 0$.
- (e) X_i iid $\text{Gamma}(\theta, 1/\theta)$, $\theta > 0$.

The gamma density uses the shape–scale convention

$$f(x; \alpha, \beta) = \{\Gamma(\alpha)\beta^\alpha\}^{-1} x^{\alpha-1} e^{-x/\beta}.$$

Solution:

- (a) Set $\mu = x_1$ and let $\sigma^2 \downarrow 0$. The first normal density diverges while the second remains positive and finite. The likelihood is unbounded, so no MLE exists.
- (b) The parameter space is $\mathbb{R} \setminus \{0\}$. Let

$$A = \sum_{i=1}^n x_i^2, \quad B = \sum_{i=1}^n x_i.$$

Then

$$\ell(\theta) = C - n \log |\theta| - \frac{A}{2\theta^2} + \frac{B}{\theta} - \frac{n}{2}.$$

The score equation is

$$n\theta^2 + B\theta - A = 0,$$

with roots

$$\theta_{\pm} = \frac{-B \pm \sqrt{B^2 + 4nA}}{2n}.$$

Fix $A > 0$; if $A = 0$ then all $x_i = 0$ and $\ell(\theta) \rightarrow \infty$ as $\theta \rightarrow 0$, so no MLE exists. On $(0, \infty)$ and on $(-\infty, 0)$, as $\theta \rightarrow 0$ the term $-A/(2\theta^2) \rightarrow -\infty$ dominates $-n \log |\theta| \rightarrow \infty$, so $\ell(\theta) \rightarrow -\infty$; as $|\theta| \rightarrow \infty$, $-n \log |\theta| \rightarrow -\infty$ while the remaining variable terms vanish or stay bounded, so again $\ell(\theta) \rightarrow -\infty$. Hence on each half-line ℓ is continuous, tends to $-\infty$ at both ends, and so attains its supremum at an interior critical point. Since $\theta_+ \theta_- = -A/n < 0$, exactly one root lies in $(0, \infty)$ and one in $(-\infty, 0)$, so θ_+ and θ_- are the (unique, hence global) maximizers of ℓ on their respective half-lines.

To compare the two, note that at any root of $n\theta^2 + B\theta - A = 0$ we have $A/\theta^2 = n + B/\theta$, so

$$\ell(\theta) = C - n - n \log |\theta| + \frac{B}{2\theta}.$$

Thus

$$\ell(\theta_+) - \ell(\theta_-) = -n(\log \theta_+ - \log |\theta_-|) + \frac{B}{2} \left(\frac{1}{\theta_+} - \frac{1}{\theta_-} \right).$$

By Vieta's formulas, $\theta_+ + \theta_- = -B/n$, i.e. $\theta_+ - |\theta_-| = -B/n$, so $\theta_+ < |\theta_-|$ exactly when $B > 0$, making the first term positive exactly when $B > 0$ (negative when $B < 0$, zero when $B = 0$). Also $\theta_+\theta_- = -A/n < 0$ gives $\frac{1}{\theta_+} - \frac{1}{\theta_-} = \frac{\theta_- - \theta_+}{\theta_+\theta_-} > 0$ always, so the second term shares the sign of B as well. Hence

$$\ell(\theta_+) > \ell(\theta_-) \iff B > 0, \quad \ell(\theta_+) = \ell(\theta_-) \iff B = 0,$$

so the global MLE is θ_+ when $B > 0$, θ_- when $B < 0$, and both roots tie for the global maximum when $B = 0$.

(c) Here

$$\ell(\theta) = C - \frac{n}{2} \log \theta - \frac{A}{2\theta} + B - \frac{n\theta}{2}.$$

The score equation is

$$n\theta^2 + n\theta - A = 0.$$

If $A > 0$, the unique positive root is

$$\hat{\theta} = \frac{-1 + \sqrt{1 + 4A/n}}{2},$$

and it is the unique MLE. If $A = 0$, the likelihood is unbounded as $\theta \downarrow 0$, so no MLE exists.

(d) Let

$$S = \sum_{i=1}^n x_i, \quad G = \sum_{i=1}^n \log x_i.$$

Then

$$\ell(\theta) = C + (\theta - 1)G - \frac{S}{\theta} - n \log \Gamma(\theta) - n\theta \log \theta.$$

Writing ψ and ψ_1 for the digamma and trigamma functions, the score and its derivative are

$$U(\theta) = G + \frac{S}{\theta^2} - n\psi(\theta) - n(\log \theta + 1)$$

and

$$U'(\theta) = -\frac{2S}{\theta^3} - n\psi_1(\theta) - \frac{n}{\theta} < 0.$$

Moreover, $U(\theta) \rightarrow \infty$ as $\theta \downarrow 0$ and $U(\theta) \rightarrow -\infty$ as $\theta \rightarrow \infty$. Hence the MLE is the unique positive solution of $U(\theta) = 0$.

(e) Now

$$\ell(\theta) = C + n\theta \log \theta - n \log \Gamma(\theta) + (\theta - 1)G - \theta S.$$

Thus

$$U(\theta) = n(\log \theta + 1) - n\psi(\theta) + G - S$$

and

$$U'(\theta) = n \left(\frac{1}{\theta} - \psi_1(\theta) \right) < 0.$$

Also,

$$\lim_{\theta \rightarrow \infty} U(\theta) = n + G - S = \sum_{i=1}^n (1 + \log x_i - x_i) \leq 0.$$

The inequality is strict unless every $x_i = 1$. If the data are not all equal to 1, there is a unique positive root of $U(\theta) = 0$, and it is the MLE. If every $x_i = 1$, there is no finite MLE; the likelihood approaches its supremum as $\theta \rightarrow \infty$.

9. (0 points) **Problem 1.9.**

Suppose $X_1, \dots, X_n$ are iid Bernoulli(θ). For an integer k , find the UMVUE of θ^k and determine when it does not exist.

Solution: Let

$$T = \sum_{i=1}^n X_i \sim \text{Binomial}(n, \theta).$$

Then T is complete and sufficient. For $0 \leq k \leq n$, using the falling factorial $(t)_k = t(t-1)\cdots(t-k+1)$,

$$\mathbb{E}_\theta[(T)_k] = (n)_k \theta^k.$$

Therefore

$$\hat{\theta}^k = \frac{(T)_k}{(n)_k} = \frac{\binom{T}{k}}{\binom{n}{k}}$$

is unbiased and is the UMVUE by the Lehmann–Scheffé theorem.

Every expectation of a statistic on the finite Bernoulli sample space is a polynomial in θ of degree at most n . If $k > n$, the function θ^k has degree greater than n ; if $k < 0$, it is not a polynomial. Hence no unbiased estimator exists in those cases. The UMVUE exists exactly for

$$k \in \{0, 1, \dots, n\}.$$

For $k = 0$, the estimator is the constant 1.

10. (0 points) **Problem 1.10.**

Prove that any function of a complete statistic is complete.

Solution: Let T be complete and set $V = g(T)$. Suppose a is measurable, $\mathbb{E}_\theta |a(V)| < \infty$, and

$$\mathbb{E}_\theta[a(V)] = 0$$

for every θ . Define $b(t) = a(g(t))$. Then

$$\mathbb{E}_\theta[b(T)] = \mathbb{E}_\theta[a(g(T))] = 0$$

for every θ . Since T is complete, $b(T) = 0$ almost surely for every θ . But $b(T) = a(V)$, so $a(V) = 0$ almost surely for every θ . Therefore V is complete.

11. (0 points) **Problem 1.11.**

Let $X_1, \dots, X_n$ be iid $\text{Poisson}(\lambda)$.

- (a) Show that no unbiased estimator of $|\lambda - 1|$ exists.
- (b) For $k \in \mathbb{R}$, find the UMVUE of λ^k , if it exists.
- (c) For $a > 0$, find the UMVUE of $e^{-a\lambda}$. Explain the problem when $n < a$, especially when $a = 2n$.

Solution: Let

$$T = \sum_{i=1}^n X_i \sim \text{Poisson}(n\lambda).$$

The statistic T is complete and sufficient.

- (a) Suppose an integrable unbiased estimator existed. Rao–Blackwellization would give a function $h(T)$ satisfying $\mathbb{E}_\lambda[h(T)] = |\lambda - 1|$. Then

$$e^{n\lambda} \mathbb{E}_\lambda[h(T)] = \sum_{t=0}^{\infty} h(t) \frac{(n\lambda)^t}{t!}.$$

Because $h(T)$ is integrable for every $\lambda > 0$, this power series converges absolutely on every bounded interval and defines an analytic function of λ . It would have to equal

$$e^{n\lambda} |\lambda - 1|,$$

which is not differentiable at $\lambda = 1$. This is impossible. Therefore no unbiased estimator exists.

- (b) If $k \in \{0, 1, 2, \dots\}$, then

$$\mathbb{E}_\lambda[(T)_k] = (n\lambda)^k,$$

so

$$\hat{\lambda}^k = \frac{(T)_k}{n^k}$$

is unbiased and is the UMVUE. For any other real k , the function $e^{n\lambda}\lambda^k$ does not have an ordinary power-series expansion at $\lambda = 0$. Since the expectation of any integrable function of T produces such a power series, no integrable unbiased estimator exists.

(c) The pgf of T gives

$$\mathbb{E}_\lambda[r^T] = e^{n\lambda(r-1)}.$$

Taking $r = 1 - a/n$ yields

$$\widehat{e^{-a\lambda}} = \left(1 - \frac{a}{n}\right)^T.$$

It is unbiased and, as a function of the complete sufficient statistic, is the UMVUE. When $a > n$, the base is negative, so the estimator changes sign even though the target is positive. When $a = 2n$, the estimator is $(-1)^T$ and

$$\text{Var}_\lambda((-1)^T) = 1 - e^{-4n\lambda}.$$

Thus the estimator remains unbiased but is highly variable and oscillates between -1 and 1 .

12. (0 points) **Problem 1.12.**

For a regular one-parameter family and a one-to-one reparametrization $\gamma = g(\theta)$, show that

$$I(\gamma; X) = \frac{I(\theta; X)}{\{g'(\theta)\}^2}.$$

Solution: By the chain rule,

$$U_\gamma(X) = \frac{\partial}{\partial \gamma} \log f(X; \theta(\gamma)) = U_\theta(X) \frac{d\theta}{d\gamma} = \frac{U_\theta(X)}{g'(\theta)}.$$

Squaring and taking expectations gives

$$I(\gamma; X) = \mathbb{E}[U_\gamma(X)^2] = \frac{I(\theta; X)}{\{g'(\theta)\}^2}.$$

13. (0 points) **Problem 1.13.**

For a one-dimensional exponential family in natural form, prove

$$I(\eta; X) = -d_0''(\eta).$$

In the original parameterization, prove

$$I(\theta; X) = -d''(\theta) + \frac{d'(\theta)c''(\theta)}{c'(\theta)}.$$

Solution: In natural form,

$$f(x; \eta) = \exp\{\eta T(x) + d_0(\eta) + S(x)\} I_A(x),$$

so

$$\ell''(\eta; x) = d_0''(\eta).$$

Regularity gives

$$I(\eta; X) = -\mathbb{E}_\eta[\ell''(\eta; X)] = -d_0''(\eta).$$

In the original parameterization,

$$\ell(\theta; x) = c(\theta)T(x) + d(\theta) + S(x).$$

The mean-zero score identity gives

$$c'(\theta)\mathbb{E}_\theta[T(X)] + d'(\theta) = 0,$$

so, when $c'(\theta) \neq 0$,

$$\mathbb{E}_\theta[T(X)] = -\frac{d'(\theta)}{c'(\theta)}.$$

Since

$$\ell''(\theta; x) = c''(\theta)T(x) + d''(\theta),$$

we obtain

$$I(\theta; X) = -\mathbb{E}_\theta[\ell''(\theta; X)] = -d''(\theta) + \frac{d'(\theta)c''(\theta)}{c'(\theta)}.$$

14. (0 points) **Problem 1.14.**

Let $X_1, \dots, X_n$ be iid $\mathcal{N}(\theta, \theta)$, $\theta > 0$.

- Find the Fisher information.
- Find the efficiencies of $\bar{X}$ and s^2 and determine which is better.
- Determine whether an unbiased estimator uniformly improves both.
- Determine whether an unbiased estimator achieves efficiency one.

Solution: Let

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Since $X_i \sim \mathcal{N}(\theta, \theta)$, both

$$\mathbb{E}_\theta[\bar{X}] = \theta \quad \text{and} \quad \mathbb{E}_\theta[s^2] = \theta.$$

Thus $\bar{X}$ and s^2 are both unbiased estimators of θ .

(a) For one observation,

$$X \sim \mathcal{N}(\mu(\theta), v(\theta)),$$

where

$$\mu(\theta) = \theta, \quad v(\theta) = \theta.$$

For a normal distribution with both mean and variance depending on θ ,

$$I_1(\theta) = \frac{\{\mu'(\theta)\}^2}{v(\theta)} + \frac{1}{2} \frac{\{v'(\theta)\}^2}{v(\theta)^2}.$$

Here,

$$\mu'(\theta) = 1, \quad v'(\theta) = 1,$$

so

$$I_1(\theta) = \frac{1}{\theta} + \frac{1}{2\theta^2} = \frac{2\theta + 1}{2\theta^2}.$$

Because the observations are independent,

$$I_n(\theta) = nI_1(\theta) = \frac{n(2\theta + 1)}{2\theta^2}.$$

Therefore, the Cramér–Rao lower bound for any unbiased estimator of θ is

$$\frac{1}{I_n(\theta)} = \frac{2\theta^2}{n(2\theta + 1)}.$$

(b) We have

$$\text{Var}_\theta(\bar{X}) = \frac{\theta}{n},$$

and, because $(n-1)s^2/\theta \sim \chi_{n-1}^2$,

$$\text{Var}_\theta(s^2) = \frac{2\theta^2}{n-1}.$$

Define the efficiency of an unbiased estimator δ of θ by

$$\text{Eff}(\delta) = \frac{1/I_n(\theta)}{\text{Var}_\theta(\delta)}.$$

Then

$$\text{Eff}(\bar{X}) = \frac{\frac{2\theta^2}{n(2\theta+1)}}{\frac{\theta}{n}} = \frac{2\theta}{2\theta+1},$$

and

$$\text{Eff}(s^2) = \frac{\frac{2\theta^2}{n(2\theta+1)}}{\frac{2\theta^2}{n-1}} = \frac{n-1}{n(2\theta+1)}.$$

Thus,

$$\text{Eff}(\bar{X}) > \text{Eff}(s^2)$$

if and only if

$$\frac{2\theta}{2\theta+1} > \frac{n-1}{n(2\theta+1)}.$$

Since $2\theta+1 > 0$, this is equivalent to

$$2n\theta > n-1,$$

or

$$\theta > \frac{n-1}{2n}.$$

Therefore:

$$\begin{cases} \bar{X} \text{ is more efficient,} & \theta > \frac{n-1}{2n}, \\ s^2 \text{ is more efficient,} & 0 < \theta < \frac{n-1}{2n}, \\ \bar{X} \text{ and } s^2 \text{ have equal efficiency,} & \theta = \frac{n-1}{2n}. \end{cases}$$

(c) The joint density is

$$\begin{aligned} f_{\theta}(x_1, \dots, x_n) &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi\theta}} \exp \left\{ -\frac{(x_i - \theta)^2}{2\theta} \right\} \\ &= (2\pi\theta)^{-n/2} \exp \left\{ -\frac{1}{2\theta} \sum_{i=1}^n x_i^2 + \sum_{i=1}^n x_i - \frac{n\theta}{2} \right\}. \end{aligned}$$

Thus the joint density factors through

$$T = \sum_{i=1}^n X_i^2.$$

Hence T is sufficient.

Moreover, this is a one-dimensional exponential family with natural statistic T and natural parameter

$$\eta = -\frac{1}{2\theta}, \quad \eta \in (-\infty, 0).$$

The representation is minimal (the single statistic T is non-constant on the support, so condition (b) of Problem 1.4 holds trivially with $k = 1$), and the natural parameter space $(-\infty, 0)$ is open in $\mathbb{R}$. By the completeness theorem for full-rank (minimal) exponential families with open natural parameter space, T is therefore complete sufficient.

Now define

$$\delta(T) = \mathbb{E}_\theta[\bar{X} \mid T].$$

Since T is sufficient, the conditional distribution $P_\theta(X \in \cdot \mid T = t)$ does not depend on θ ; write this common conditional law as $P(\cdot \mid t)$. Defining $\delta(t) = \mathbb{E}[\bar{X} \mid T = t]$ using this single θ -free kernel produces one function of T alone that is simultaneously a version of $\mathbb{E}_\theta[\bar{X} \mid T]$ for every θ . Since $\bar{X}$ is unbiased,

$$\mathbb{E}_\theta[\delta(T)] = \mathbb{E}_\theta[\bar{X}] = \theta.$$

Thus $\delta(T)$ is an unbiased function of the complete sufficient statistic T , and hence it is the UMVUE of θ .

Likewise,

$$\mathbb{E}_\theta[s^2 \mid T]$$

is an unbiased function of T . Since both

$$\mathbb{E}_\theta[\bar{X} \mid T] \quad \text{and} \quad \mathbb{E}_\theta[s^2 \mid T]$$

are unbiased functions of the complete sufficient statistic T , completeness implies that they are equal almost surely. Therefore the same estimator $\delta(T)$ is the Rao–Blackwell improvement of both $\bar{X}$ and s^2 .

Consequently,

$$\text{Var}_\theta(\delta(T)) \leq \text{Var}_\theta(\bar{X})$$

and

$$\text{Var}_\theta(\delta(T)) \leq \text{Var}_\theta(s^2)$$

for every $\theta > 0$. Hence an unbiased estimator uniformly improves both $\bar{X}$ and s^2 .

(d) The log likelihood, up to terms not depending on θ , is

$$\ell(\theta) = -\frac{n}{2} \log \theta - \frac{T}{2\theta} - \frac{n\theta}{2}.$$

The score is

$$U_\theta(X) = \frac{\partial \ell}{\partial \theta} = -\frac{n}{2\theta} + \frac{T}{2\theta^2} - \frac{n}{2} = \frac{T - n\theta - n\theta^2}{2\theta^2}.$$

Suppose that an unbiased estimator $\delta(X)$ attained the Cramér–Rao lower bound for every $\theta > 0$. Equality in the Cramér–Rao inequality would imply

$$\delta(X) - \theta = \frac{1}{I_n(\theta)} U_\theta(X).$$

Since

$$\frac{1}{I_n(\theta)} = \frac{2\theta^2}{n(2\theta + 1)},$$

we would obtain

$$\begin{aligned} \delta(X) - \theta &= \frac{2\theta^2}{n(2\theta + 1)} \cdot \frac{T - n\theta - n\theta^2}{2\theta^2} \\ &= \frac{T - n\theta - n\theta^2}{n(2\theta + 1)}. \end{aligned}$$

Therefore,

$$\delta(X) = \theta + \frac{T - n\theta - n\theta^2}{n(2\theta + 1)} = \frac{T + n\theta^2}{n(2\theta + 1)}.$$

The right-hand side depends nontrivially on the unknown parameter θ . Hence it cannot equal one fixed statistic $\delta(X)$ for every $\theta > 0$. Therefore no unbiased estimator attains efficiency one uniformly over the parameter space.

15. (0 points) **Problem 1.15.**

Let X_i be iid $\text{Poisson}(\lambda)$ and let $\hat{\gamma}$ be the UMVUE of $\gamma = e^\lambda$.

- (a) Find its mean and variance.
- (b) Show that $\text{Var}(\hat{\gamma}) > 1/I(\gamma; X)$.
- (c) Show that the ratio tends to one as $n \rightarrow \infty$.

Solution: Let

$$T = \sum_{i=1}^n X_i \sim \text{Poisson}(n\lambda).$$

Using the Poisson pgf,

$$\hat{\gamma} = \left(1 + \frac{1}{n}\right)^T$$

is unbiased for e^λ . Since it is a function of the complete sufficient statistic T , it is the UMVUE.

(a) The pgf gives

$$\mathbb{E}[\hat{\gamma}] = e^\lambda$$

and

$$\mathbb{E}[\hat{\gamma}^2] = \exp \left\{ n\lambda \left(\left(1 + \frac{1}{n}\right)^2 - 1 \right) \right\} = e^{2\lambda + \lambda/n}.$$

Therefore

$$\text{Var}(\hat{\gamma}) = e^{2\lambda} (e^{\lambda/n} - 1).$$

(b) Since $I(\lambda; X) = n/\lambda$ and $\gamma = e^\lambda$, the reparametrization formula gives

$$I(\gamma; X) = \frac{n}{\lambda e^{2\lambda}}, \quad \frac{1}{I(\gamma; X)} = \frac{\lambda e^{2\lambda}}{n}.$$

Because $e^x - 1 > x$ for $x > 0$,

$$e^{2\lambda} (e^{\lambda/n} - 1) > \frac{\lambda e^{2\lambda}}{n}.$$

Hence $\text{Var}(\hat{\gamma}) > 1/I(\gamma; X)$.

(c) The ratio is

$$\frac{\text{Var}(\hat{\gamma})}{1/I(\gamma; X)} = \frac{e^{\lambda/n} - 1}{\lambda/n} \longrightarrow 1 \quad (n \rightarrow \infty).$$

16. (0 points) **Problem 1.16.** (Optional)

If $T(X) \in \mathbb{R}^k$ is sufficient and there exist $\theta_0, \dots, \theta_k$ such that

$$\left(\frac{f(x; \theta_1)}{f(x; \theta_0)}, \dots, \frac{f(x; \theta_k)}{f(x; \theta_0)} \right)$$

is a one-to-one function of $T(x)$, prove that T is minimal sufficient.

Solution: Work on the common support where the displayed ratios are defined. Since T is sufficient, the factorization theorem implies that

$$T(x) = T(y) \implies \frac{f(x; \theta)}{f(y; \theta)} \text{ is independent of } \theta.$$

Conversely, suppose $f(x; \theta)/f(y; \theta)$ is independent of θ . Then for $i = 1, \dots, k$,

$$\frac{f(x; \theta_i)/f(x; \theta_0)}{f(y; \theta_i)/f(y; \theta_0)} = 1.$$

Hence the two ratio vectors at x and y are equal. By the assumed one-to-one relationship with T , this implies $T(x) = T(y)$. Therefore

$$T(x) = T(y) \iff \frac{f(x; \theta)}{f(y; \theta)} \text{ is independent of } \theta.$$

The likelihood-ratio characterization of minimal sufficiency now shows that T is minimal sufficient.

17. (0 points) **Problem 1.17.**

If $X \sim \text{Gamma}(\alpha, \beta)$ in the shape–scale parameterization, find $\mathbb{E}_{\alpha, \beta}[\log X]$, $\text{Var}_{\alpha, \beta}(\log X)$, and $\text{Cov}_{\alpha, \beta}(\log X, X)$ using the exponential-family representation.

Solution: The density is

$$f(x; \alpha, \beta) = \exp \left\{ (\alpha - 1) \log x - \frac{x}{\beta} - \log \Gamma(\alpha) - \alpha \log \beta \right\} I_{(0, \infty)}(x).$$

Take

$$T_1(X) = \log X, \quad T_2(X) = X, \quad \eta_1 = \alpha - 1, \quad \eta_2 = -\frac{1}{\beta}.$$

The log-partition function is

$$A(\eta) = \log \Gamma(\eta_1 + 1) - (\eta_1 + 1) \log(-\eta_2).$$

Using $\nabla A(\eta) = \mathbb{E}_\eta[T(X)]$ (and $\mathbb{E}_\eta[T(X)] = \mathbb{E}_{\alpha, \beta}[T(X)]$ under the change of variables $\eta \leftrightarrow (\alpha, \beta)$),

$$\mathbb{E}_{\alpha, \beta}[\log X] = \frac{\partial A}{\partial \eta_1} = \psi(\eta_1 + 1) - \log(-\eta_2) = \psi(\alpha) + \log \beta.$$

Using $\nabla^2 A(\eta) = \text{Cov}_\eta(T(X)) = \text{Cov}_{\alpha, \beta}(T(X))$,

$$\text{Var}_{\alpha, \beta}(\log X) = \frac{\partial^2 A}{\partial \eta_1^2} = \psi_1(\alpha)$$

and

$$\text{Cov}_{\alpha, \beta}(\log X, X) = \frac{\partial^2 A}{\partial \eta_1 \partial \eta_2} = -\frac{1}{\eta_2} = \beta.$$

18. (0 points) **Problem 1.18.**

Let $X = (X_1, \dots, X_n)$ be iid $\text{Gamma}(\alpha, \beta)$. Prove or disprove: if

$$\mathbb{E}_{\alpha, \beta}[g(X)] = 0$$

for all $\alpha, \beta > 0$, then

$$\mathbb{E}_{\alpha, \beta} \left[g(X) \mid \sum_{i=1}^n X_i, \prod_{i=1}^n X_i \right] = 0$$

for all $\alpha, \beta > 0$.

Solution: The assertion is true, assuming $g(X)$ is integrable, i.e. $\mathbb{E}_{\alpha, \beta}|g(X)| < \infty$, at every (α, β) with $\alpha, \beta > 0$ (this is implicit in the hypothesis $\mathbb{E}_{\alpha, \beta}[g(X)] = 0$ for all $\alpha, \beta > 0$, and is needed for the conditional expectations below to be well defined).

The joint density is

$$\begin{aligned} f_{\alpha, \beta}(x) &= \prod_{i=1}^n \frac{1}{\Gamma(\alpha)\beta^\alpha} x_i^{\alpha-1} e^{-x_i/\beta} I_{(0, \infty)}(x_i) \\ &= \exp \left\{ (\alpha - 1) \sum_{i=1}^n \log x_i - \frac{1}{\beta} \sum_{i=1}^n x_i - n \log \Gamma(\alpha) - n\alpha \log \beta \right\} I_{(0, \infty)^n}(x). \end{aligned}$$

Therefore,

$$T_1(X) = \sum_{i=1}^n X_i, \quad T_2(X) = \sum_{i=1}^n \log X_i$$

is a sufficient statistic. Its natural parameters are

$$\eta_1 = \alpha - 1, \quad \eta_2 = -\frac{1}{\beta},$$

with natural parameter space

$$(-1, \infty) \times (-\infty, 0),$$

which is open in $\mathbb{R}^2$.

Also, the representation is minimal: the statistics 1, T_1 , and T_2 are linearly independent. Hence, by the full-rank exponential-family completeness theorem,

$$(T_1, T_2) = \left(\sum_{i=1}^n X_i, \sum_{i=1}^n \log X_i \right)$$

is complete and sufficient.

Since

$$\sum_{i=1}^n \log X_i = \log \left(\prod_{i=1}^n X_i \right),$$

the two statistics

$$\left(\sum_{i=1}^n X_i, \sum_{i=1}^n \log X_i \right)$$

and

$$\left(\sum_{i=1}^n X_i, \prod_{i=1}^n X_i \right)$$

generate the same sigma-field.

By sufficiency, there exists a parameter-free version

$$h(T_1, T_2) = \mathbb{E}_{\alpha, \beta}[g(X) \mid T_1, T_2].$$

By the tower property,

$$\mathbb{E}_{\alpha, \beta}[h(T_1, T_2)] = \mathbb{E}_{\alpha, \beta}[g(X)] = 0$$

for every $\alpha, \beta > 0$. Since (T_1, T_2) is complete,

$$h(T_1, T_2) = 0$$

almost surely for every $\alpha, \beta > 0$.

Therefore,

$$\mathbb{E}_{\alpha, \beta} \left[g(X) \mid \sum_{i=1}^n X_i, \prod_{i=1}^n X_i \right] = 0$$

almost surely for all $\alpha, \beta > 0$.

19. (0 points) **Problem 1.19.**

Suppose $Y_1, \dots, Y_n$ are independent with

$$Y_i \sim \mathcal{N}(\beta x_i, \sigma^2),$$

where the x_i are fixed and known.

- (a) Find complete sufficient statistics.
- (b) Find the MLEs of (β, σ) .
- (c) Find the UMVUEs of β and σ^2 .

Solution: Let

$$q = \sum_{i=1}^n x_i^2 > 0, \quad T_1 = \sum_{i=1}^n x_i Y_i, \quad T_2 = \sum_{i=1}^n Y_i^2.$$

(a) Expanding the joint density gives

$$f_{\beta, \sigma^2}(y) = (2\pi)^{-n/2} \exp \left\{ \frac{\beta}{\sigma^2} T_1 - \frac{1}{2\sigma^2} T_2 - \frac{\beta^2 q}{2\sigma^2} - \frac{n}{2} \log \sigma^2 \right\}.$$

Thus (T_1, T_2) is sufficient. The natural parameters are

$$\eta_1 = \frac{\beta}{\sigma^2}, \quad \eta_2 = -\frac{1}{2\sigma^2},$$

and they range over the open set $\mathbb{R} \times (-\infty, 0)$.

The design is assumed nondegenerate, i.e. not every x_i is 0, so that $q = \sum_{i=1}^n x_i^2 > 0$; this is exactly the condition needed for minimality. Indeed, by the criterion of Problem 1.4, minimality requires that $b_1 T_1(y) + b_2 T_2(y) = b_0$ for all $y \in \mathbb{R}^n$ forces $b_0 = b_1 = b_2 = 0$. Comparing the coefficient of y_i^2 gives $b_2 = 0$; comparing the coefficient of y_i then gives $b_1 x_i = 0$ for every i , and since $q > 0$ some $x_i \neq 0$, so $b_1 = 0$; finally $b_0 = 0$. Hence the representation is minimal (for every $n \geq 1$; we additionally take $n \geq 2$ below so that the UMVUE of σ^2 in part (c) has positive degrees of freedom). Hence the full-rank exponential-family completeness theorem shows that (T_1, T_2) is complete sufficient.

(b) Least squares gives

$$\hat{\beta} = \frac{T_1}{q}.$$

The residual sum of squares is

$$\text{RSS} = \sum_{i=1}^n (Y_i - \hat{\beta} x_i)^2 = T_2 - \frac{T_1^2}{q}.$$

Therefore, when $\text{RSS} > 0$,

$$\hat{\sigma}_{\text{MLE}}^2 = \frac{\text{RSS}}{n}, \quad \hat{\sigma}_{\text{MLE}} = \sqrt{\frac{\text{RSS}}{n}}.$$

If $\text{RSS} = 0$, the likelihood is unbounded as $\sigma^2 \downarrow 0$, so no MLE exists.

(c) The estimator $\hat{\beta} = T_1/q$ is unbiased and is a function of the complete sufficient statistic. Therefore

$$\hat{\beta}_{\text{UMVUE}} = \frac{\sum_{i=1}^n x_i Y_i}{\sum_{i=1}^n x_i^2}.$$

For $n \geq 2$,

$$\frac{\text{RSS}}{\sigma^2} \sim \chi_{n-1}^2,$$

so $\mathbb{E}[\text{RSS}] = (n-1)\sigma^2$. Hence

$$\hat{\sigma}_{\text{UMVUE}}^2 = \frac{\text{RSS}}{n-1} = \frac{1}{n-1} \left(T_2 - \frac{T_1^2}{q} \right).$$

Both estimators are functions of the complete sufficient statistic, so the Lehmann–Scheffé theorem proves the UMVUE claims.